

Exam 1

Lauren Hubert

1. Some of the beneficial safety features of SFRs:
- a) Negative reactivity feedback - negative $\alpha +$ means reactivity decreases w/ an increase in temperature
 - b) Na has high thermal conductivity
 - c) Sodium and stainless steel are very compatible
 - d) Long thermal transient times
 - e) Large margin to boiling
 - f) Can operate at atmospheric pressure, since at these temperatures Na is a liquid.

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2. Alloying uranium in fuels has different benefits depending on what it is alloyed with. Alloying with smaller atoms creates sinks for point defects. This helps to reduce swelling. Alloying at higher content allows lower the solidus line so that more of the Uranium is in gamma phase. This is beneficial for isotropic, predictable swelling behavior. Alloying with Zr reduces swelling, stabilizes the BCC (gamma) phase, and increases the melting temp of the fuel.

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phase transition $\uparrow$
actually $\uparrow$ solidus

3. metallic fuel systems have a high amount of swelling, leading to FCMI. The solution to this to prevent interaction between the fuel and the cladding is to include more free space to accommodate the swelling. The swelling is anisotropic due to the hotter center being more gamma phase, with larger porosity. This leads to more radial swelling. The alpha phase uranium also swells anisotropically. This is because of the large thermal spikes caused by irradiation-induced displacement that leads to interstitial and vacancy loops forming on certain planes. The best way to limit FCMI is with more free space.

- plenum?

4. Constituent redistribution in metallic fuels is referring to the way that different elements move around within the fuel. This is due to different temperature gradients and chemical gradients driving diffusion. The Zr especially is redistributed around due to thermal effects and solvent diffusion kinetic effects. The solubility of Zr varies with which phase of Uranium it is in, leading to diffusion along isotherms. The solid fission products in metallic fuel also diffuse and collect at the fuel/cladding boundary, leading to FCCI in that zone. Different solid fission products precipitate out differently; some dissolve into the bond sodium, some precipitate as separate compound, and some help counteract swelling by taking the place of U or Pu as they fission out. The complex phases in the fuel leads to complex solubility gradients, as well as chemical and thermal gradients, that all drives constituent redistribution in metallic fuels.

- redistribution is really only the primary constituents (U, Pu, Zr)

5. As burnup increases in metallic fuels, thermal conductivity decreases as porosity develops. The gaseous fission products create swelling and voids that degrade thermal conductivity. Then, as these bubbles grow, they vent into the plenum. This creates an opening for the hot sodium to enter. The hot sodium entering increases the thermal conductivity for the rest of life.

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6. FCCI is a localized interdiffusion zone between the fuel and cladding, which leads to weakening of the cladding and a lower melting temperature of the fuel. In this local diffusion zone, there is also grain boundary diffusion occurring. The primary driver is the fission products which diffuse down the thermal gradient and collect along the cladding. They attack the cladding, and Fe and Cr start to diffuse into the fuel region. The fuel species participating are Pu and the solid fission products that precipitate out, like Cs. Impurities present may form a Zr-rich rind that helps to prevent the interdiffusion. FCCI is the main problem and life-limiting phenomena of metallic fuels. The complex thermodynamic and chemical gradients are hard to predict with models, and we do not have the diffusion constants we need for all the different phases present. The main failure point is the cladding hoop strain.

Nd + Ce main two FP_s

7. Restructuring in MOX fuel happens because of the high temperatures and strong thermal gradient. It leads to a large central void, a columnar growth region, equiaxed region, and an as-sintered region. Pores form with a cold and hot side and the condensation/evaporation makes them to collect at the center, leaving columnar growth behind and forming the central void. MOX fuels can operate at high temperatures, but have lower thermal conductivity.

- why columnar / equiaxed switch?

8. Pu moves up the thermal gradient in MOX fuels, because the VO_2 is more volatile and the condensation/evaporation of the pores makes it collect in the center of the fuel. O does the opposite, and diffuses down the thermal gradient to collect on the fuel/cladding interface. This is what leads to JOG forming here. The M/O ratio varies from 2.0 at the center to 1.03 at the interface because of this. During manufacturing, Pu agglomerates may also have been formed, creating local high fission hot-spots.

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9. ZOG is a porous, squishy oxide layer that forms locally on the fuel/cladding interface. It has "good" thermal conductivity properties and can help with FCCI. It forms from the oxygen produced during fission increasing the oxygen potential, diffusing down the thermal gradient, and collecting on the cladding interface. It is bad for FCCI because of the fission products the oxide contains that increase interdiffusion between fuel & cladding. ZOG stands for Joint Oxide Growth. We try to have a hypostoichiometric MO at beginning of life to limit ZOG.

10. Type I: sodium dissolution
Type II: impurities interacting w/ sodium

The main driver for sodium corrosion in SFR's is thermal gradient mass transfer. In the core, high temperature allow more dissolution of steels, especially Ni, and carry the particulates down to cooler parts of the system. Hence, the solubility of the elements (Fe, Cr, Ni) are lower in Na so they plate out into the piping downstream of the IHX.

This corrosion mechanism does not saturate, and has been observed through mass loss/mass gain in the components.

The velocity of Na, decarburization/carburization, temperature, irradiation time, presence of oxygen are all factors in the corrosion mechanisms of SFRs.

The Na itself does not have detrimental effects, but has effects that weaken materials and cause corrosion.

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